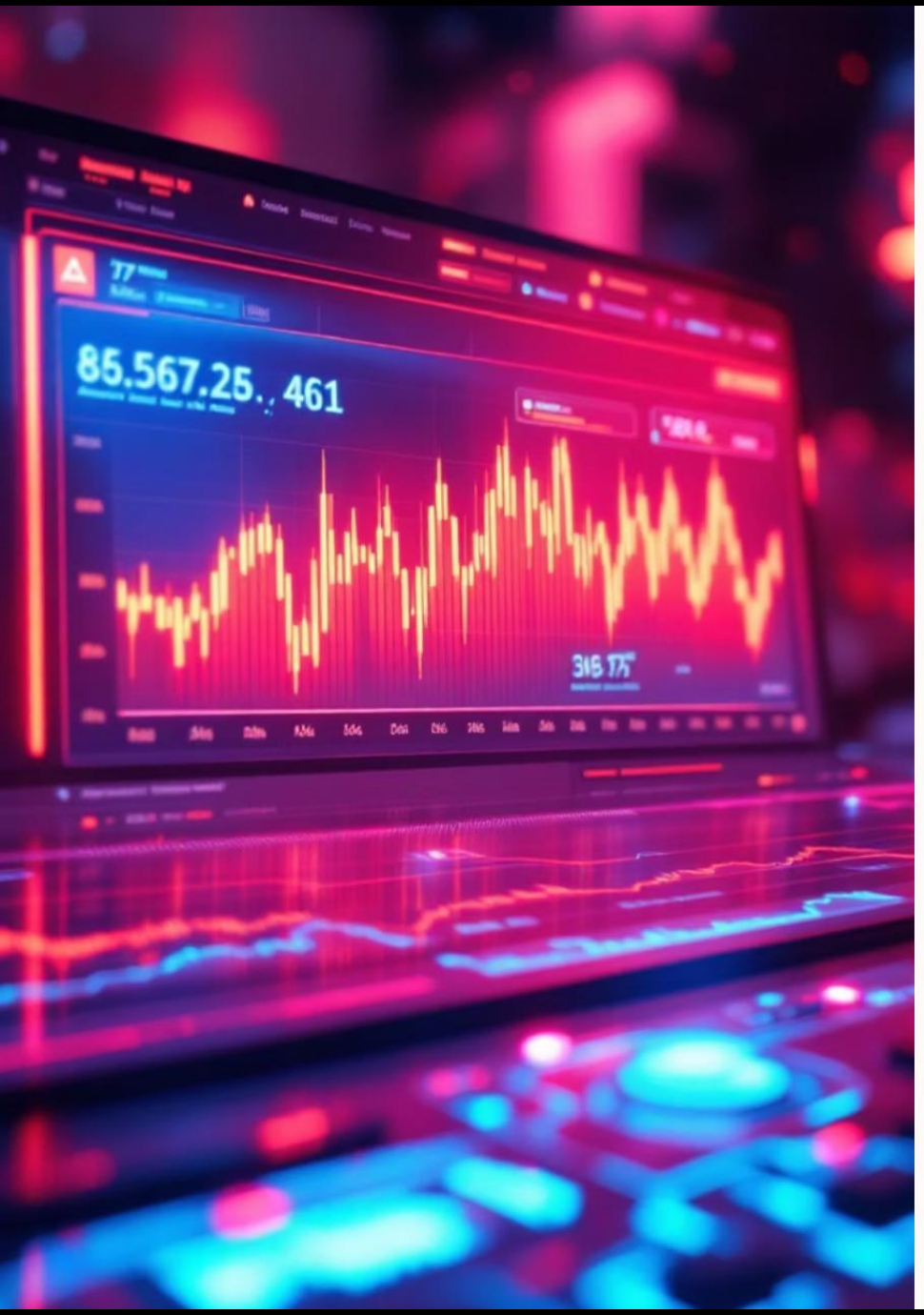


Market Liquidity Stress Detection

Order Book Microstructure Analysis on 1-Minute BTC Data

BY Omkar Kadam

Executive Summary

Core Objective

Design a Liquidity Stress Detection System using 1-minute Bitcoin microstructure data to monitor risk in real-time.

Composite Index

Quantify cross-sectional stress levels using a composite Liquidity Stress Index (LSI).

Market States

Categorize conditions into Normal, Warning, and Critical regimes for actionable insights.

The Business Problem

Flash Liquidity Droughts

High-frequency crypto environments suffer from instant liquidity evaporation due to systemic shocks or structural imbalances.

Key Risks

- Execution fragility with significant slippage
- Risk contagion through widening spreads
- Institutional hedging challenges

Dataset Overview

17,113	156	1-min
Total Observations	Features	Granularity
Rows of high-granularity data	Microstructure-related columns	Bitcoin order book data

Key Metrics	Midpoint price, Bid-ask spread, Buy/Sell volume, Order book depth levels
Time Period	April 7-19, 2021

Feature Engineering

01

Spread-Based Liquidity Shock

Calculate Z-score deviation from 60-minute rolling mean to identify abnormal spreads.

02

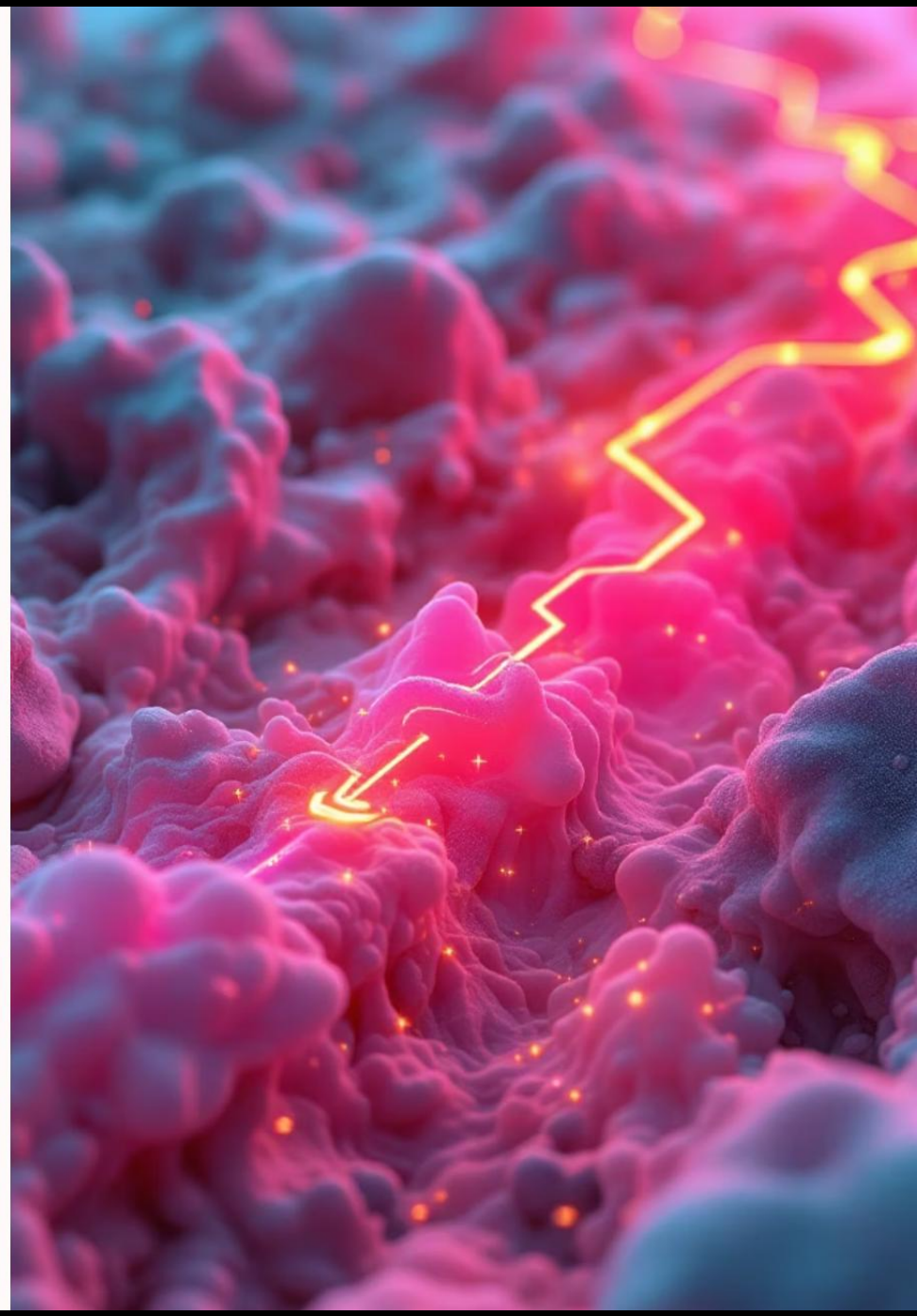
Order Flow Imbalance

Capture directional pressure indicating liquidity drainage on one side.

03

Depth Imbalance

Sum top 5 bid and ask levels to measure order book symmetry.



The Liquidity Stress Index

A composite score designed to rise when abnormal spreads, excessive volatility, and depth imbalance converge.

$$LSI = \frac{1}{5}\sum_{i=1}^{5} \text{Standardized Signal}_i$$

1

Standardize Signals

Normalize all components

2

Align Direction

Make absolute values

3

Combine Scores

Average into LSI

Stress Regime Classification

Normal

12,469 minutes

Standard market-making conditions with healthy liquidity.

Warning

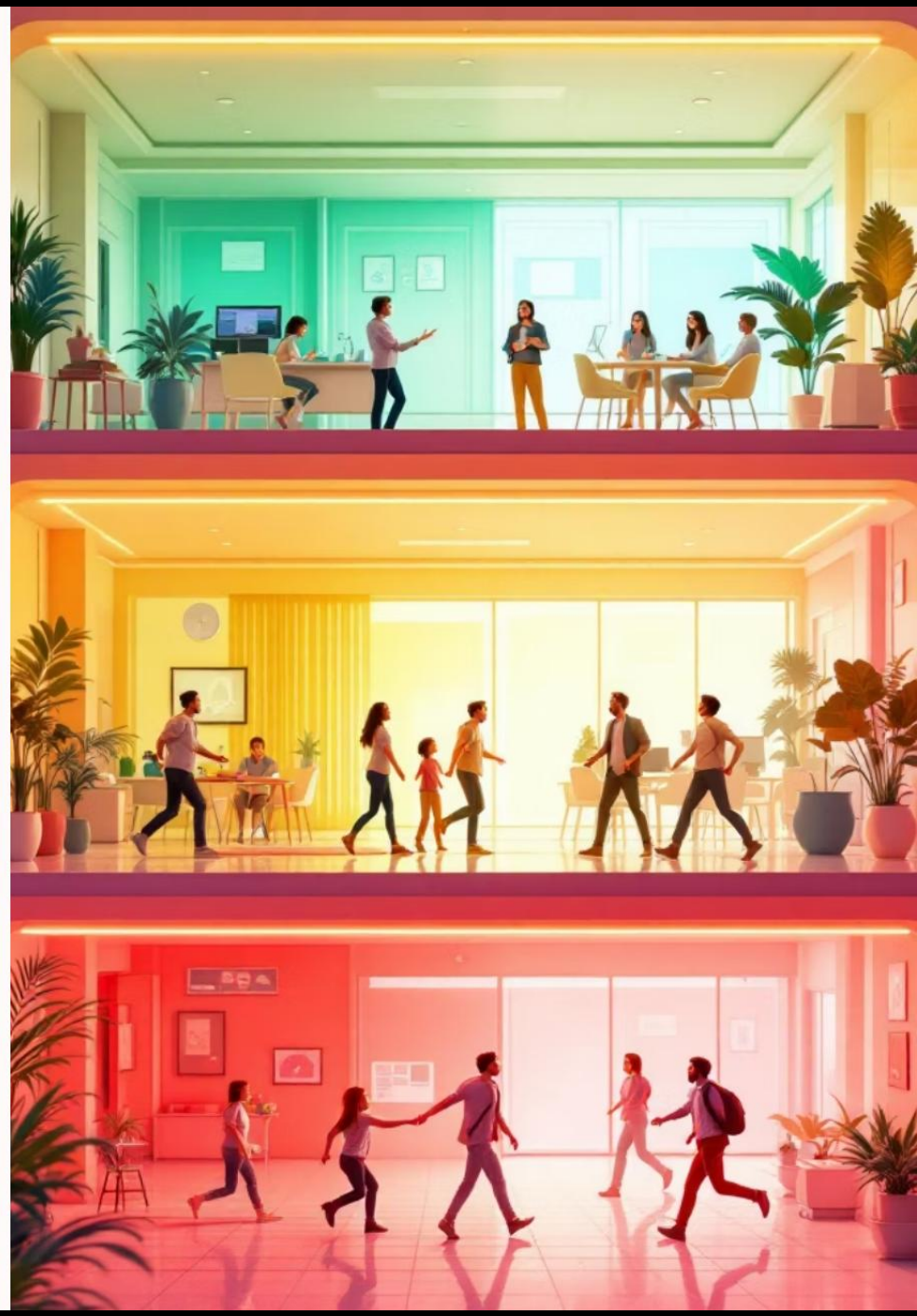
4,427 minutes

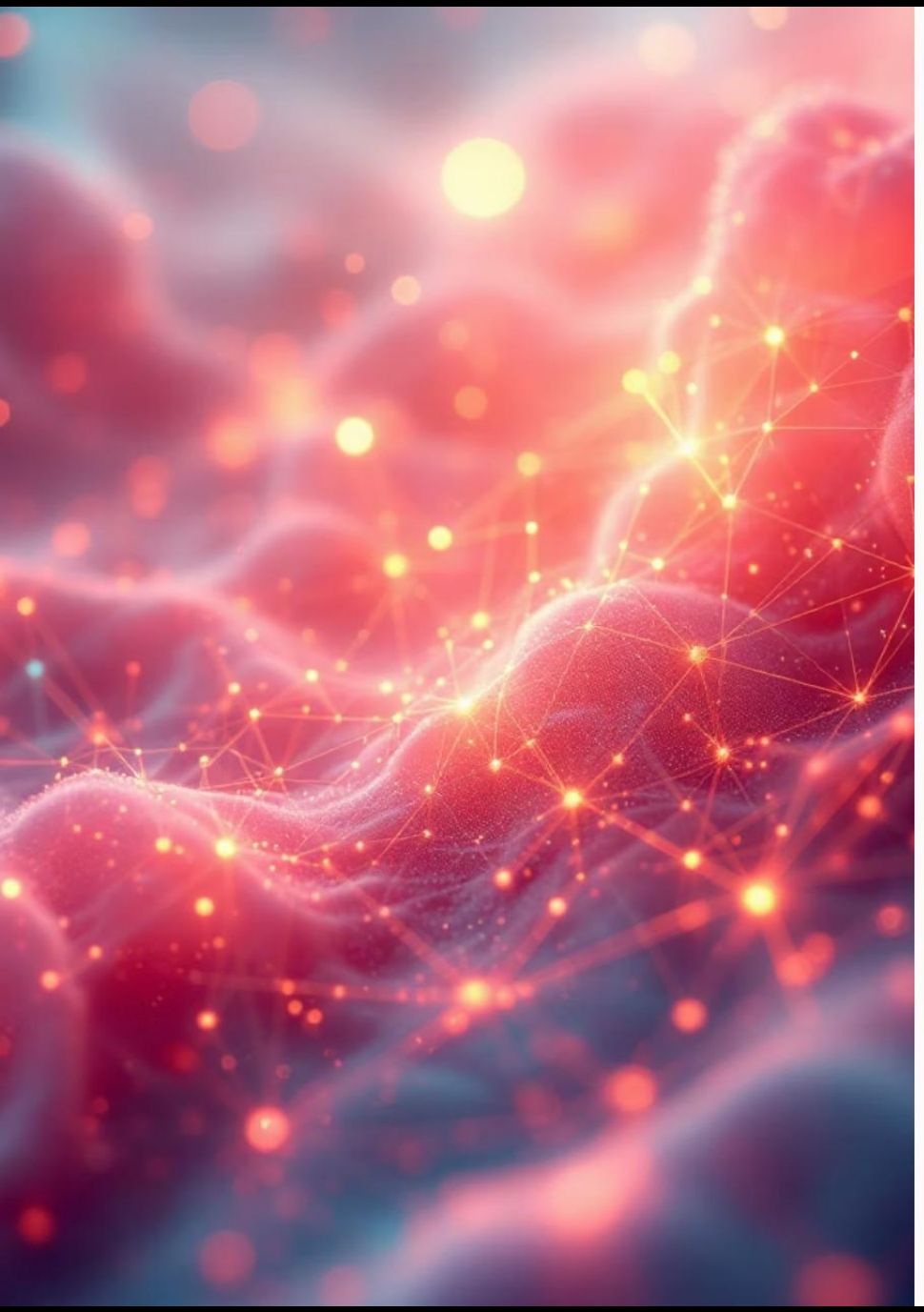
Deteriorating depth requiring caution and monitoring.

Critical

217 minutes

Extreme conditions triggering risk-off protocols.





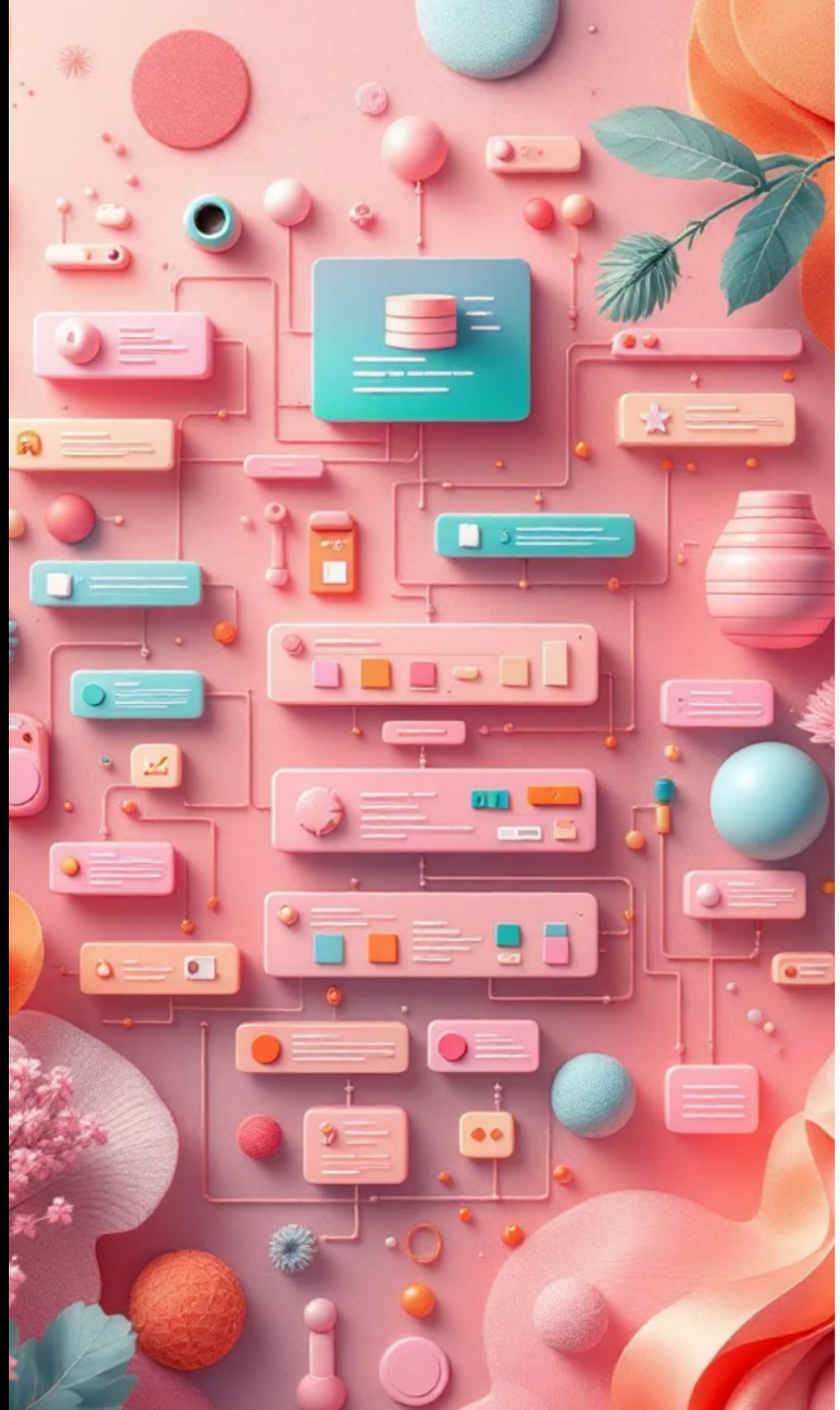
Key Insights

Clustering Behavior

Liquidity stress appears in clusters rather than random single-minute events, providing a window to act before total breakdown.

Volatility Relationship

While immediate predictive correlation is limited ($r \approx -0.013$), critical regimes consistently exhibit highest average volatility.



Production Integration

1

Daily Summaries

Snapshot of liquidity health per day stored in MySQL database for reporting.

2

Regime Aggregations

Percentage of time spent in Critical status for trend analysis.

3

Real-Time Pipeline

Simulates production data warehouse for desk traders and risk systems.

Conclusion & Future Roadmap

Current Applications

HFT risk controls, slippage estimation models, exchange monitoring dashboards.

Future Enhancements

ML regime classification, multi-asset validation (ETH, SOL), API real-time alerts.

Skills Demonstrated

Python, Pandas, NumPy, rolling statistics, feature engineering, anomaly detection, SQL, risk scoring framework design.

This project successfully bridges raw market microstructure data and actionable risk intelligence, proving systemic fragility can be quantified and managed.